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A COMPARATIVE ANALYSIS OF ARCHITECTURAL EXPERIENCE IN THE REALITY CONTINUUM BASED ON EMBODIMENT, TACTILITY AND AFFORDANCE

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Abstract. This paper presents a comparative analysis between a physical model, a set of virtual reality (VR) models and a set of augmented reality (AR) models of the same architectural project. A set of architectural participants' interactions with the models have been recorded through a set of sensor streams and videos and their experiences have been interpreted through a questionnaire containing both qualitative and quantitative questions. Further analysing through the sensor streams, video recording and questionnaire, their experiences have been interpreted using a framework of embodiment, affordance and tactility to determine the usefulness and limitations and each of the modes and their possible application in architectural design practice.

Keywords. Augmented Reality, Virtual Reality, Architectural Model, Embodied Cognition, Affordance, Spatial Cognition

1. Introduction

Historically, in the architectural design process, two-dimensional drawings, physical models, and materials have played a crucial role in visualising and experiencing the space before it was built(Mindrup & Wells, 2020). The role played by the physical model in this paradigm was important and preferred by many artists, architects, and patrons throughout history for various structural and experiential properties. Architects like Callimachus of Greece during the Hellenic period, Michelangelo and the Medicis during the Renaissance, Antoni Gaudi during art nouveau, Frei Otto during Modernism and Alvar Alto, Peter Zumthor, Rem Koolhas in the post-modern to present day have explored the physical materiality and spatial qualities with physical models of varying typology and scale. In the past 30 years, with the development of technology, computer-aided design (CAD), 3D modelling capabilities, building information modelling (BIM) and photorealistic visualizations have grown and have developed on an unprecedented scale. Consequently, these

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developments have majorly influenced the architectural design process in many ways. The efficiency and ease of 3D CAD and BIM models have many ways replaced the practice of using a physical model in many architectural practices. Additionally, using immersive virtual reality (IVR) many architectural practices have almost entirely replaced the architectural model except for the final presentation stage from a material, cost and time involvement perspective while still obtaining valuable spatial and architectural insights(AIA, 2006). The physical and virtual reality models reside on two extremes in the reality continuum(Fig. 1) (Milgram & Kishino, 1994). Hence, the question arises if these VR models can entirely replace the role of physical models or if different modes of models can still contribute anything specific to the design process.

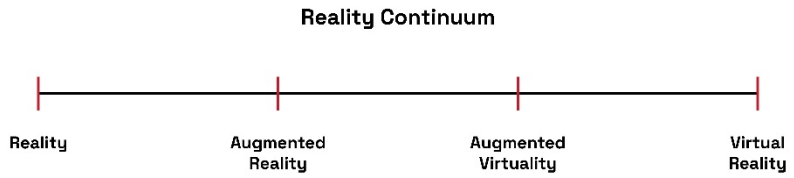


Figure 1:Reality Continuum, Drawn after Milgram & Kishino 1994

Previous studies have been conducted by Sun et al to determine if the spatial reasoning capabilities are similar between a computer graphics model seen on a screen and a physical model of the same (Sun et al., 2013). The study almost entirely focused on estimating dimensional perception between the models but did not offer much insight into issues of embodiment and tactile perception as the participants were not allowed to touch the models. Though the study concluded by mentioning that the physical model was better at height and distance estimation, it is imperative to mention that the other setup was not an immersive virtual reality but a computer screen-based virtual reality where movements were restricted. Later in 2018, Hermund et al studied the perception of an architectural atmosphere through VR, physical space itself and 2D representation of the space through plan and section (Hermund et al., 2018). In their study, VR simulates similar nature of perception of being in the actual space while 2D drawings like plans and sections and photorealistic non-immersive visualizations are not as effective as the former one.

In the past 4 years due to the rapid development of inside-out tracking technology, consumer-grade standalone augmented reality (AR) and virtual reality (VR) devices have become available which can be used in a smaller space through internal tracking. Similarly, ARKit and ARCore for iOS and Android devices have brought AR to mobile devices democratizing these mixed reality (MR) applications. Also, in physical manufacturing, rapid prototyping technologies have reduced the fabrication time for 3D objects. A combination of these technological advancements has helped to explore the different modalities along the reality continuum in a much-simplified manner.

In this project, we have developed four basic modalities of one architectural model: a physical model; through an immersive VR headset (Meta Quest pro); a handheld AR device (iPad Pro) and through a head-mounted AR headset

(HoloLens2). Participants with architectural education were selected to interact with the model with a follow-up questionnaire about the spatial configuration of the architectural model they have interacted with. Through both qualitative and quantitative analysis of the data obtained from the participants, the study will fundamentally investigate two key interests: (1) how different degrees of embodiment affect the scale and spatial perception of the space through different modalities and (2) how affordances of these modalities influence the interaction with the model and in turn how it affects our perception of the space. Based on our research interest we would like to put forward the hypothesis: Spatial perception offered by the different modalities over the reality continuum are different in nature and should not be ideally interchanged for the sake of ease of use. In this paper, we will present the effect of embodiment, affordances, and tactility on the understanding of the spatial perception and spatial cognition of the user by studying the participants' degree and manner of interaction with the model. Finally, the paper will identify the unique role of each modality over the human perception of architectural space.

2. Methods

French Philosopher Maurice Merleau-Ponty, in his seminal work, represented the "body" not only from the sole anatomical perspective or a static passive entity but as an entry point to explore reality; the body as a tool with certain affordances that helps us experience our immediate world(Hale, 2017). In this case, the body acts as a compass of navigation in our immediate world, be it virtual or real. On a very similar note, for the ocular-centric approach to architectural experience, renaissance master Michelangelo often used the metaphor "seste negli occhi" or compass in the eyes emphasizing the importance of appearance and the degree of realism that encompassed architectural representation(Conforti et al., 2020). Through these ideas "body as a compass" and "eye as a compass" we would like to like to explore the importance of movement, visual quality, and embodiment in spatial cognition.

In this experiment, participants needed to experiment with a single architectural project in different modalities. An architectural project was chosen among three shortlisted projects showing enough spatial complexity and consisting of volumes of different scales. In this case, it is a model of an institutional building made for a cultural icon. Physical and mesh models were made of the same in 1:200 scale of similar material appearance. In this case, the material was chosen to be white material without any resemblance to any physical material that might have been used on the building. Regarding the level of detail, this model could be classified as a LOD 300 model.

2.1. PARTICIPANT SELECTION

This experiment aims to implement the results in the discourse architectural design practice. Thus, the participants chosen for the experiment have a minimum of five years of architectural training in academia and a basic understanding of mixed-reality tools. The experience of the subjects varies from two years after graduation to twenty-five years of experience. The participants Also, in terms of experience with MR tools, there is a good mix of amateurs to experts among the participants. All the participants

are from academia with a range of research assistants, assistant and associate professors and PhD fellows.

2.2. ARCHITECTURAL PROJECT

The architectural project used in this experiment is an unbuilt competition entry of a memorial for a cultural icon from Square Consultancy Services, Kolkata. The model has been used with their permission. The project can be classified as a cultural building which encompasses a stairway to the main floor and a spiralling volume with large balconies towards an artificial water body.

2.3. PHYSICAL MODEL

The physical model was made using laser-cut white cardboard and 3D-printed parts. The laser cut burns were kept in the model as a distinct nature rather than covered up through paint to retain the appearance of an architectural model. The model was also fitted with an inertial measurement unit and a data logger to record the movements in terms of rotation around the X, Y and Z axis while handled during the experiments. The physical model has a dimension of 420mm x 400mm x 60 mm on a 1:200 scale. It was also fitted with an HTC Vive Tracker which helped to corroborate the IMU data using external tracking.

2.4. VR & AR APPLICATION

A CAD model was made in Rhino3D which was consequently converted to a closed-mesh model to export in VR and AR applications. For the VR application, Meta Quest Pro was used in the experiment as it became available during the process and the colour pass-through mode helped decide a certain aspect of the experiments. Arkio (Arkio - Collaborative Spatial Design, 2022;) was used as the VR application to facilitate the viewing of the model in 1:1 and 1:200 scale. Arkio also offers a passthrough mode through which users can see the model in their surroundings, mimicking augmented reality while retaining the same graphic quality.

AR application was divided into two subcategories: handheld and head-mounted. For handheld AR, an iPad Pro with a 13-inch screen was used while Hololens2 was used for the head-mounted option. For both cases, Fologram(Fologram, 2021) was used to connect and facilitate the AR experience as well as track users' movement. The choice of iPad was added to the mix due to the price point of the Hololens2. To avoid discrepancy in experience it was decided to advise the participants not to move the AR model as the mode of interaction was different in two devices: one was through the touchscreen, and another was through spatial grabbing/pinching using hands.

During the experiment, a set of sensor streams was captured from the physical model and head-mounted displays and iPad. After the experiment, each participant had to answer a questionnaire about their experience with the models in different modalities. In this experiment, only commercially available and consumer-grade hardware, software, and applications were used to collect data from the participants.

3. Design Experimentation

Though the visual perception of the space appears as the primary stimuli or the gateway to experience the architecture it is a multisensory and tactile discourse(Pallasmaa, 2012). Spatial volumes alone can evoke certain emotions but our bodily interactions like walking, touching, and smelling can append/modify our base understanding of the space. This sensory-motor coupling helps us create a better understanding of the world(Hostetter & Alibali, 2008). Often through computer graphics and visualization, we tend to compress this multisensory experience into an ocular-centric experience which is essentially the case of immersive virtual reality or augmented reality-based representation.

In this design experiment, the participants have experienced one space in different modalities which are spread over the reality continuum encompassing a different level of embodied cognition. Each participant in this experiment went through 6 different modes of representations of the space: physical model in 1:200 scale, VR model in 1:200 scale, 3D models in 1:200 scale with passthrough enabled which has the controls and visual qualities of Arkio but seems like an AR, VR model in 1:1 scale, AR model through Hololens2 and AR model through an iPad. For each mode participants got two minutes to explore the model once they became comfortable with navigation in that mode. In each participant's case, the order of the modalities was shuffled to ensure a non-biased opinion on the experience itself. To ensure the participant can navigate the model space, three control questions were asked at each stage; if they can clearly see the model, if they can navigate with ease and if they feel at ease with the modality and the environment; once they were comfortable the experiment continued. The experiment was conducted in a studio space where students usually work with physical models and projects with which the participants are familiar. This choice was made to ensure the experiment space should not hinder the movement of the participants. After each participant went through all six modalities, a questionnaire was presented to the participants to record their experience which had both qualitative and quantitative parameters.

4. Results

The results of the experiment gave insight into how different modalities can offer different experiences on the same model. According to the post-experiment questionnaire, 71% of preferred the 1:200 scale model as the most favourable exploratory medium while 29% preferred the VR 1:1 model. But as a second preference, 66% chose VR 1:1 model. On the other hand, 71% of participants ranked

Hololens2-based AR as the least preferred option and 29% mentioned iPad-based AR owing to mostly their translucent and "fragile" style of visual representation (see Fig. 2). Further insights are classified into three categories: (1) movement in and around the model, (2) embodied cognition/bodily perception and (3) scale perception.

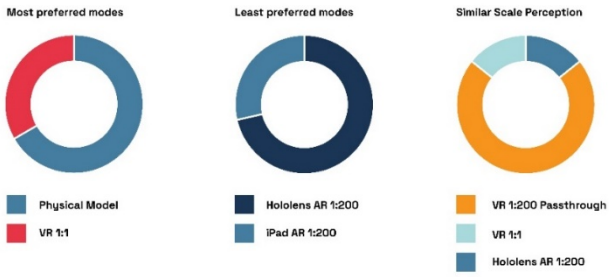


Figure 2: Modality preference and Scale perception

4.1. MOVEMENT & EMBODIED COGNITION

For all the modalities, the headsets and handheld devices were tracked through either inside-out tracking data or outside-in tracking. The finding is explained in the following sections.

When we look at a 3D object naturally, we hold it in our hand or move around the object based on the affordances. In the case of the physical model, it has been observed that everyone picked up the model and mostly rotated the model around the Z axis. In comparison, the movement around the X and Y axis is quite minimal (Fig. 3). This data is mainly obtained from the IMU fitted with the model and the tracked path through HTC Tracker is also in tandem with this observation.

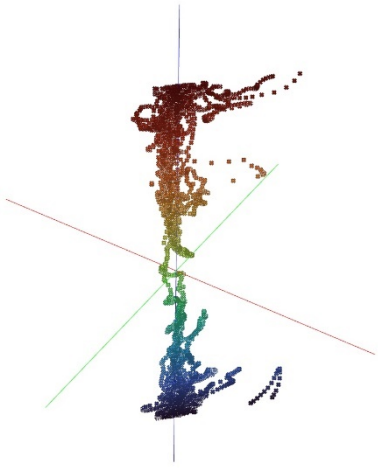


Figure 3: Rotation Data Distribution of Physical Model

A similar observation was found in the case of AR-based visualizations but in this case, participants chose to walk around the model as picking up in the model was not a viable solution, but the patterns of the movement are different in nature. In the case of Hololens2-based AR, participants mostly walked around in mostly circular pattern while looking at the model from a distance (Fig. 4).

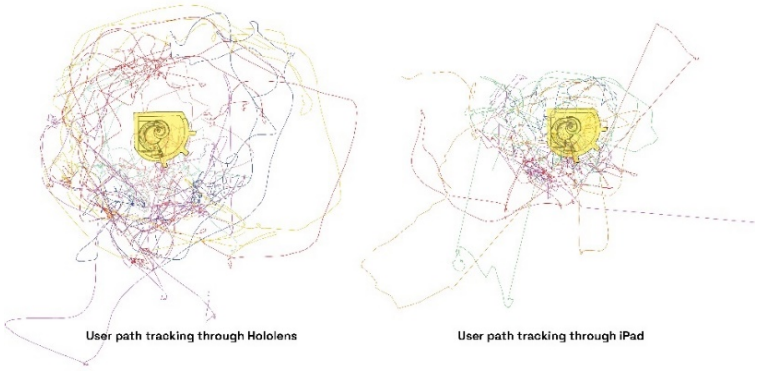


Figure 4: User Movement tracking through AR Devices

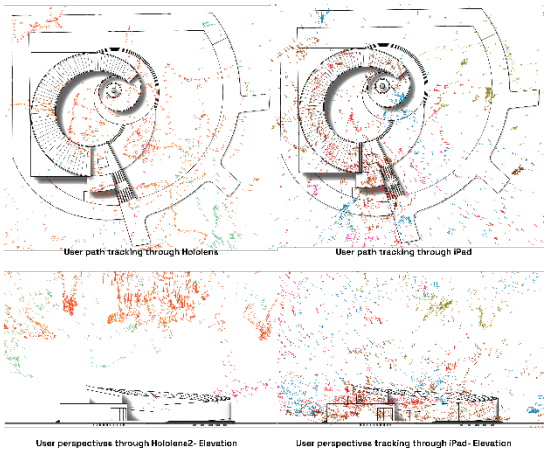


Figure 5: User Perspective and Extent of User reach in AR

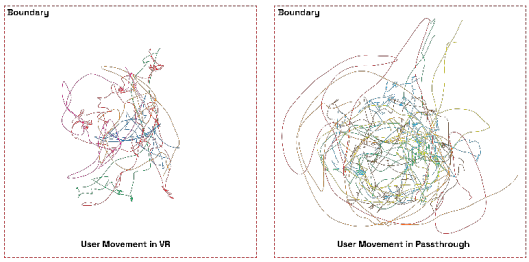
The narrow field of view of Hololens2 could be a contributing factor to this behaviour. In contrast to this observation, in iPad-based AR, participants moved more freely while looking at the model and due to the affordance, they could look inside the model (See Fig. 5). The handheld nature of the device gave the participants a unique perspective on the model and the ability to control the movement. Also, participants reported, the attachment to the device made them less aware of their bodies and the affordance of the iPad extended to the body which limited the body

movement.

In the case of VR 1:200, the observation was like the physical model; participants picked up the model and put it up at eye level while rotating the model itself. But in the passthrough mode, participants were observed to move around more around the model as they were made aware of their surroundings (See Fig. 6). This phenomenon happened despite the participants knowing they were within a safe boundary, and they could be warned if they wanted to leave the same.

But in the case of VR 1:1, participants observed some difficulties as they could not walk around a large model or climb up a stair in the model. In the 1:1 mode, the participants need to "teleport" themselves in the model using the joystick, which resembles jumping movement in the real world and deters the participants from exploring the model compared to the other modes. Often in this mode participants found themselves lost in the model. This effect is known to be present in the VR and causes disorientation of the participant in the space as the participant is not using their body to move around causing a disjunction between sensory and motor function(Prithul et al., 2021).

Figure 6: User Movement in VR & Passthrough



4.2. SCALE PERCEPTION

As humans, we usually do not look at one object in isolation but create a set of relationships with the object and its immediate surroundings(Berger, 2008). Participants were asked if we consider the 1:200 physical model as a benchmark then which modality was the closest in scale compared with the physical model. Here again, 72% of the participants mentioned VR 1:200 in passthrough mode was the closest in the scale while 14% Hololens2 visualization was the most similar in scale while the rest of the participants mentioned that iPad AR was closest in scale comparison. Their close relationship with the surroundings in those modes could be a driving factor for that result. In VR 1:1, participants' scale perception in one single perspective is reported to be well while understanding spatial configuration is reported not well due to the limitation in movement. This could stem from participants' inability to map the space using their own stride which could have given a mental map of the space. Also, in the case of the physical model, the tactile perception and grabbing the model with their own body reportedly gave the participants a better understanding of the scale which was totally absent in the AR-based visualizations but faintly present in VR-based ones due to the VR controller.

5. Discussion

In this experiment, it has firmly been ascertained that each of the modalities has its benefits and limitations. The agency of the physical model is well established in the architectural educational discourse which helps architects lean towards the physical model given a choice. But VR is being used in most practices for its versatility in scales and advantages over quick visualization, like how rapid prototyping disrupted the manufacturing industry. But the limitation of movements in the VR environment even in the commercially available tools sometimes hinders the purpose of the tool. Due to technical development in the last two years, VR headsets have leapt forward towards modes like passthrough which allows the user to the ground reality while seeing the model in the device. These modes enable the user to be "present" their reality and still view the models in another reality thus ranking higher on embodied cognition. But again, the lack of haptic interaction with the model makes the physical model much more viable for a quick study without the technological overload.

Though it has been found that AR is not being used in architectural offices(Das et al., 2022) as actively as VR and physical models, their utility through low-cost handheld devices could offer certain pertinent insight. These devices can be used to view the model from the outside as well as can help to look inside the model exposing the spatial configurations. But the visualization style for the AR visualizations needs to be improved by a certain degree to make it a viable alternative.

Based on the evidence presented it could be indicated that due to classical architectural training, many architects might prefer the physical model but to explore the space in detail, AR models could pose a significant opportunity for architects using low-cost handheld devices. On the other hand, through VR one can study the volumetric quality of the space better than a physical model or a screen. These technologies are part of a spectrum, and their utility should be complemented by one another based on the goals and relevant stages of architectural design, pertinent to the advantages and affordances they provide.

Some technological development is focused on developing haptic interaction devices for VR but another intermediate stage between the user and the object could again create the distance between the object and the user. But similarly, if these realities could overlap and inform one another in the architectural discourse, e.g., data visualization over the physical model like pre-visualization of unbuilt works on-site, hypothetically the acceptance of the technology could be increased by addressing the value creation and haptic capability. In all these studies, material properties and material representation play a lesser role due to technological limitations and an understanding of the digital representation of materials. The graphical representation of the models in computer graphics always reminds of a "finished" object while some physical materials like cardboard or foam give a sense of "unfinished"(Rowe & Bechthold, 2017). If this material representation can be obtained in mixed reality, then the acceptance of the technology could be arguably higher.

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